Unit: 6 Unit 8: Quadratics
Topic: Graphing Quadratics and Properties of Parabolas

Name: ________________
Date: ___________

Foundation (Jalapeno)

Q1. The company's profit P is modeled by $P = -2x^2 + 12x - 5$, where x is the number of products sold. What is the axis of symmetry?
(A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$

Q2. A stock's price S is given by $S = x^2 - 4x + 10$, where x is the day number. What is the minimum price?
(A) 6
(B) 8
(C) 10
(D) 12
(E) 14

Q3. The growth of an investment A is modeled by $A = 3x^2 + 2x + 1$, where x is the year. What is the vertex?
(A) $(-1/3, 2/3)$
(B) $(-2/3, 3/2)$
(C) $(0, 1)$
(D) $(1/3, 4/3)$
(E) $(2/3, 5/3)$

Q4. A company's revenue R is given by $R = -x^2 + 6x + 2$, where x is the number of employees. What is the maximum revenue?
(A) 10
(B) 12
(C) 14
(D) 16
(E) 18

Q5. The cost C of producing x items is $C = 2x^2 + 3x + 1$. What is the y-intercept?
(A) $(0, 1)$
(B) $(0, 2)$
(C) $(0, 3)$
(D) $(1, 0)$
(E) $(2, 0)$

Q6. A tax T is calculated by $T = x^2 - 2x - 3$, where x is the income. What is the axis of symmetry?
(A) $x = -1$
(B) $x = 0$
(C) $x = 1$
(D) $x = 2$
(E) $x = 3$

Q7. A savings account's balance B is given by $B = -2x^2 + 4x + 10$, where x is the month. What is the maximum balance?
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16

Q8. The demand D for a product is modeled by $D = x^2 - 4x + 3$, where x is the price. What is the vertex?
(A) $(-2, 7)$
(B) $(2, -1)$
(C) $(2, 1)$
(D) $(4, 3)$
(E) $(6, 3)$

Open Ended Questions (FRQ)

Q9. Graph the parabola $y = x^2 - 2x - 3$. Identify the vertex, axis of symmetry, x-intercepts, y-intercept, and direction of opening. Show all work and calculations.

Q10. A business's profit P is given by $P = -x^2 + 8x - 5$. Find the maximum profit, and the number of items sold when this occurs. Show all work and calculations, including a graph of the parabola.

Chef's Tips

- Read each question carefully before answering.
- Use a calculator only when necessary.
- Show all work and calculations for free response questions.